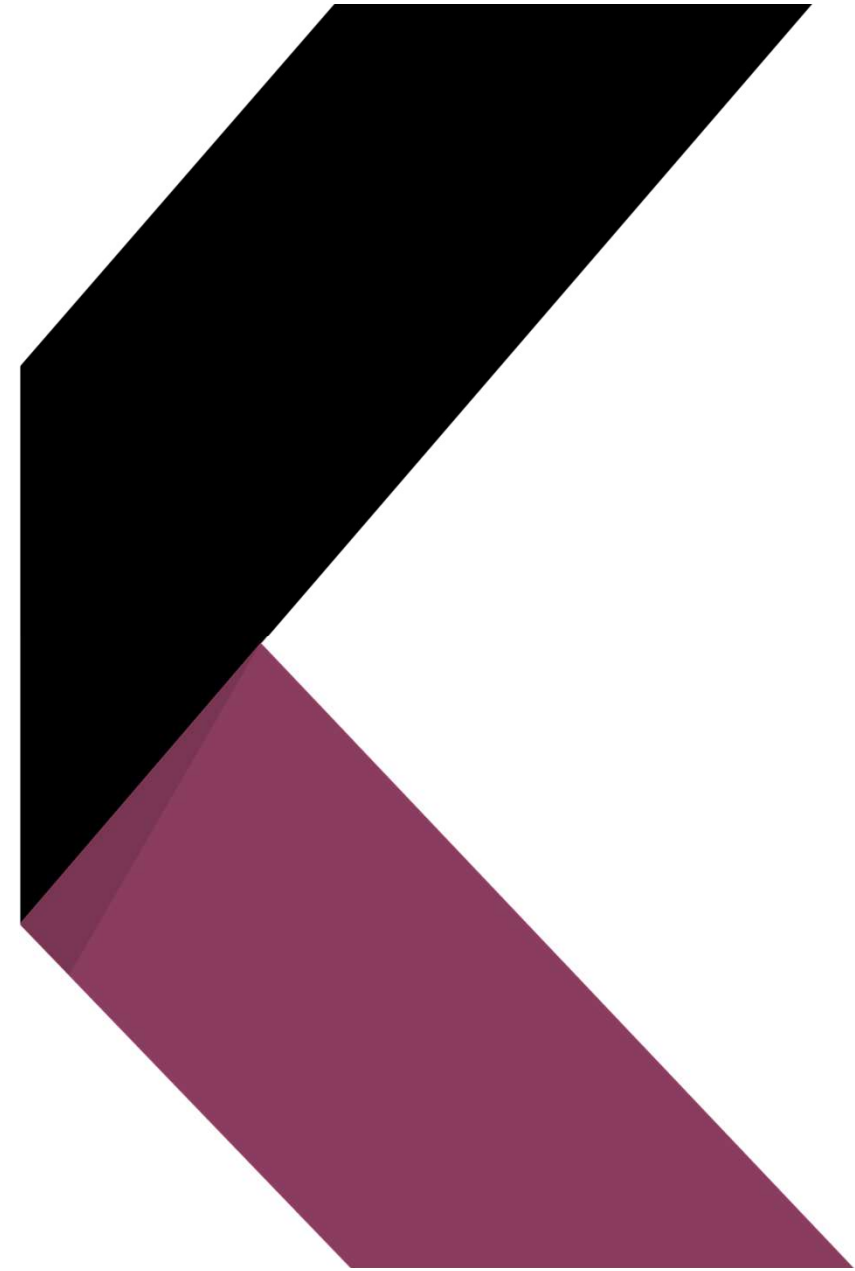


Diagnosis Process

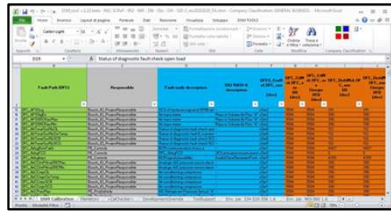
Validation



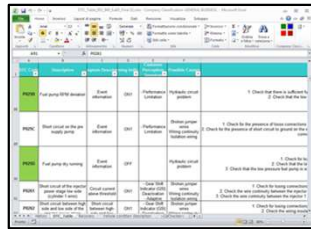


Diagnosis Process

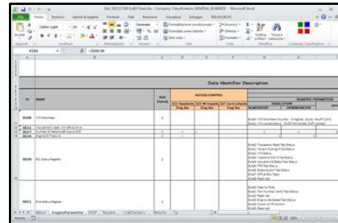
Diagnostic Validation Process



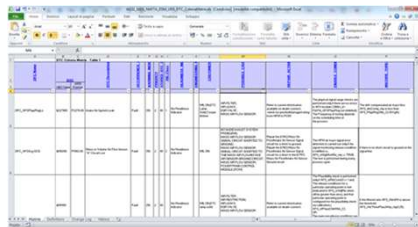
DSM



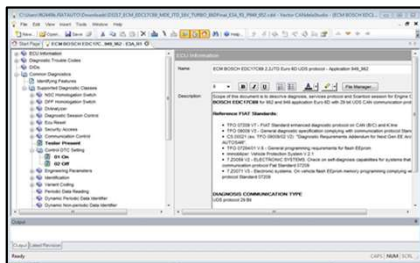
DTC Table



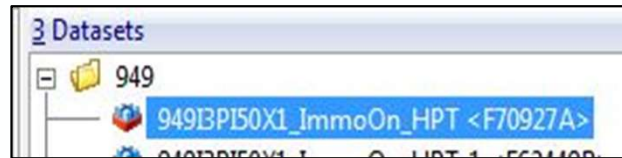
D&C



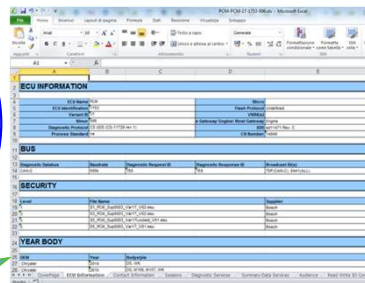
Criteria Matrix



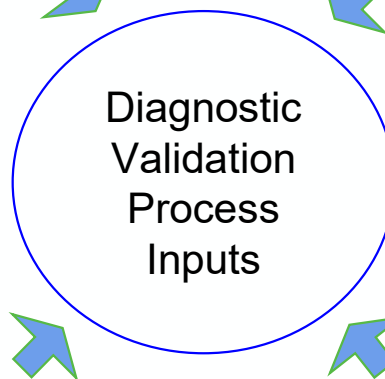
CDD



DDT

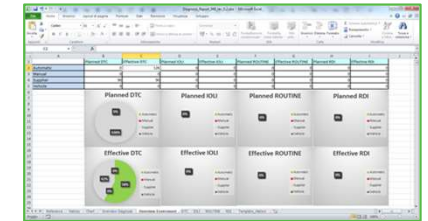


Dataset



DSM: Diagnostic System Management Application
DTC Table/Criteria Matrix: Diagnostic Trouble Codes Table
D&C: Diagnosis and Contents
CDD: Candela Diagnostic Data
DDT: Diagnostic Definition Table
Dataset: Complete SW and Calibrations for flashing
RDI: Engineering parameters
IOLI, Routine: Procedures used by Plant and Services

Reference Norms:
7.Z0059
7.Z0059/01
CS.00051
CS.00052
CS.00053



Diagnostic Report Output

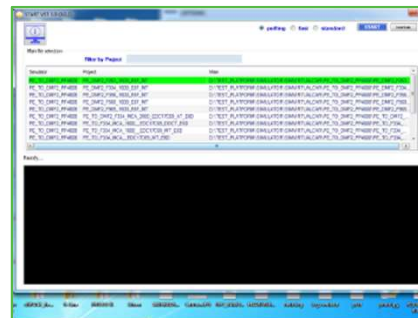
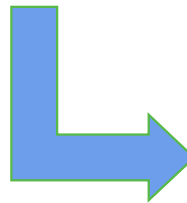
- Per project specific norms and input documents
- Coherence check between documents
- Out of spec issues tracked for discussion (DVPM)
- Report for tests done with results and acquisitions

Elements Under Test
DTC, RDI, IOLI, Routine

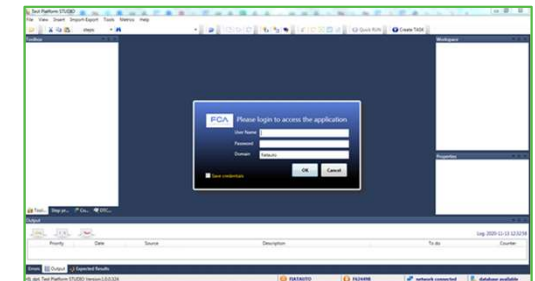
Diagnostic Validation Process - Test bench



- Tests performed manually on HIL test bench in different vehicle condition (Power On, Propulsion active, Vehicle Running) with real ECM/HCP/TCM and actuators
- Manual tests performed using ControlDesk, Inca, DIAnalyzer, CDA as per project specification
- Manoeuvres for new diagnosis are collected and shared to be automated in VST
- Tests performed by VST test platform for diagnosis already automated



VST test platform



DTC test procedure

DSM/DDT

Criteria Matrix

Checks for coherence

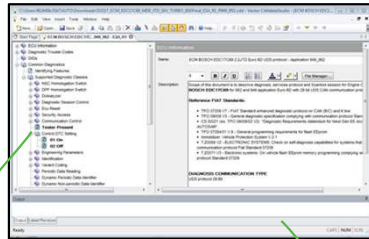
- Identify all details for the DTC
 - MIL class
 - Maturation and de-maturation time
 - Enabling and healing conditions
- Check for coherence between diagnostic documents

- RDIs in Criteria Matrix have to be active in dataset
- MIL status across documents have to be coherent
- Check all enabling and healing conditions
- Remove each condition and analyze the effect on maturation and de-maturation time
- DTC has to be stored in error memory according with the evolution of the status byte, as per project specific norms
- Check that expected recoveries work properly
- Check that unwanted recoveries are not active
- Check snapshots of DTC stored and compare RDIs of with values from other Nodes

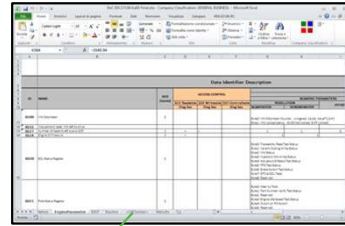


For final approval, maneuvers are done simulating all input conditions, **NOT** changing diagnosis boundary conditions

IOLI test procedure



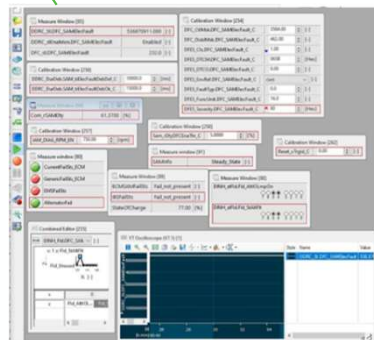
CDD/DDT



D&C

Checks for coherence

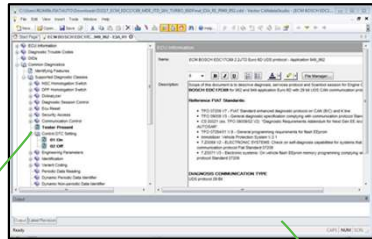
- Check the coherence between the D&C and the CDD.
- Verify that all sessions reported in D&C and in the CDD are the same and there is coherence between descriptions on both documents.



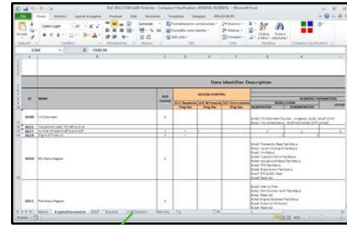
- Check all enabling conditions
- Check that the actuator under test is commanded properly
- Check all positive and negative responses of the IOLI through the diagnostic tester
- Check for correct zeroing of internal measuring points and the reset of adaptive learning parameters.
- Check for activation time as per norms



Routine test procedure



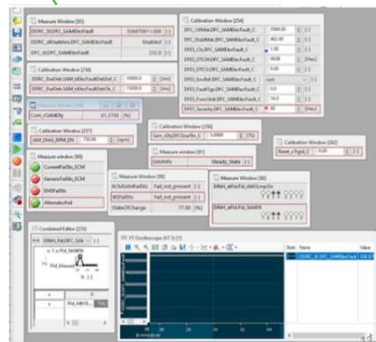
CDD/DDT



D&C

Checks for coherence

- Check the coherence between the D&C and the CDD/DDT.
- Verify that all sessions reported in D&C and in the CDD/DDT are active and coherent

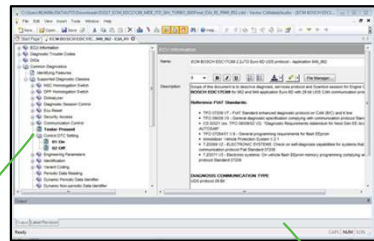


- Check all enabling conditions to start the routine command
- Check that removing them one by one the routine will answer negatively to the start command
- Check sequence for Start/Stop/Reselt Request routine commands.
- Start the routine in all diagnostic sessions and report if there are unexpected positive or negative answers

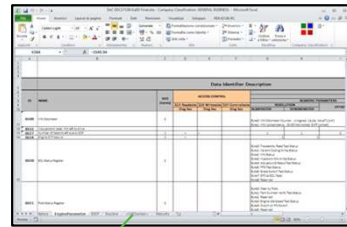


- Check the correctness of the commands for the actuators
- Check the internal zeroing measurement parameter for adaptive learning parameters
- Check for inhibiting DTC for the routine

RDI test procedure

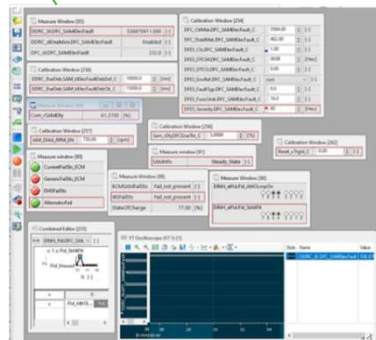


CDD/DDT



D&C

Checks for coherence



- Check the coherence between the D&C and the CDD/DDT.
- Verify that all sessions and conversion formulas reported in D&C and in the CDD/DDT are active and coherent

- Check the measuring points indicated by the D&C and CDD
- Check for all environmental conditions useful to check the behaviour of the RDI under test
- Check all enabling conditions to allow the reading and the writing of the RDI
- Read and write the RDI in all diagnostic sessions and in all addressing type
- Test the conversion formulas and perform maneuvers in order to set and unset bits of the RDI

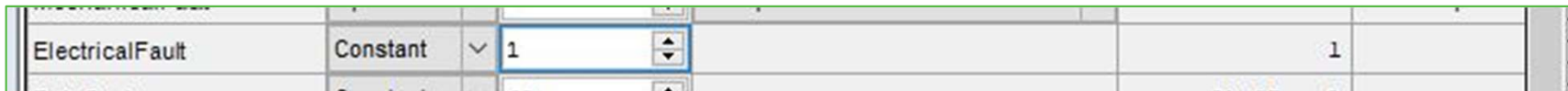
DTC test procedure 1/13



Try to simulate the conditions to set/heal the DTC in all vehicle conditions: Power On, Cranking, Engine Running, Vehicle Running and check that the vehicle conditions stated in the DTC Table are satisfied

DFC	DTC Code	DTC Description	Maturation Failure Threshold	Warning Indicator
DFC_SAMElecFault	P065B	Intelligent Alternator Module (functional test) general electrical failure (circuit short to battery or open)	Error is recognized when signal IAM_ECM_FEEDBACK.ElectricalFault = 1	OFF

Simulate the fault condition while Engine is running



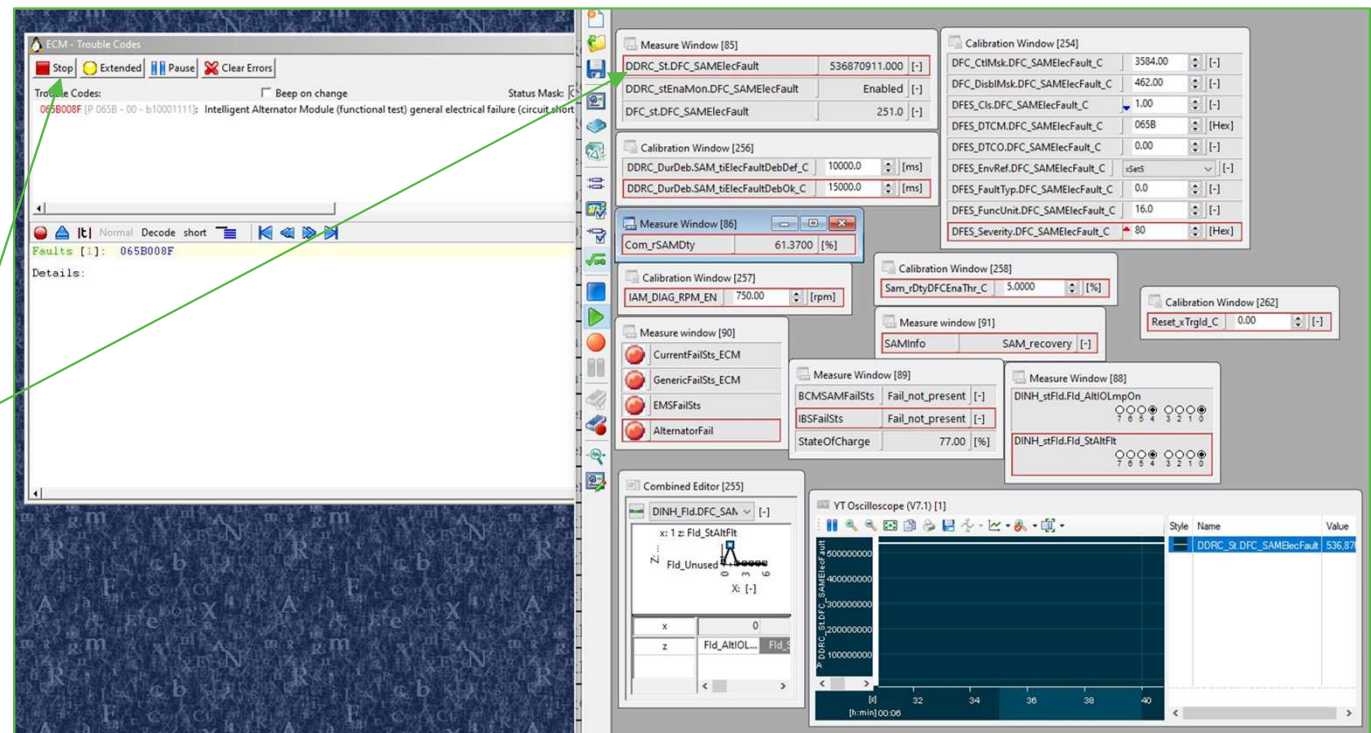
DTC test procedure 2/13

DTC under test in Engine running condition

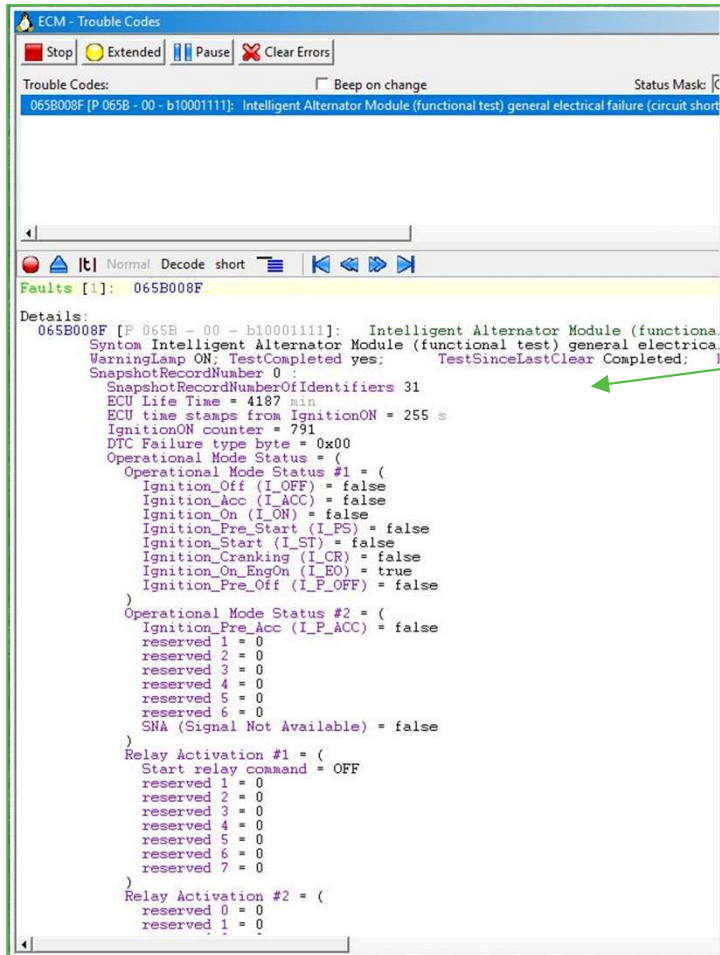
After the maturation time has elapsed, the DTC will be stored in the Error memory: Status byte is 8F

DTC Table required this fault as MIL OFF, instead the MIL is ON

Open a point on the OPL management system and ask for clarification!



DTC test procedure 3/13



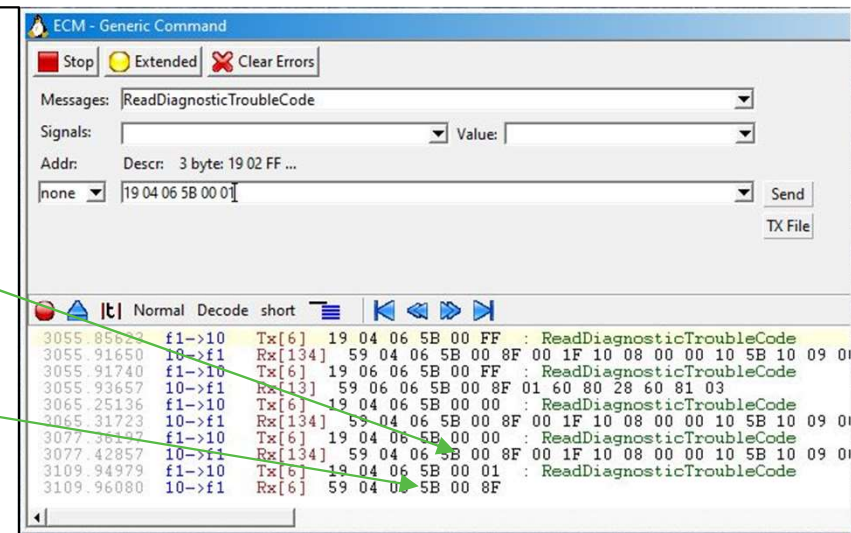
In DIAnalyzer check the coherence of the environmental parameters data

Check the values reported with the same parameters read in INCA
Environmental parameters data has NOT to contain **unknown RDI**

With service \$19 check that there is only 1 snapshot in the error memory (since it is the first time the DTC has been stored):

Send 19 04 06 5B 00 00 and check that snapshot is populated

Send 19 04 06 5B 00 01 and check that snapshot is void

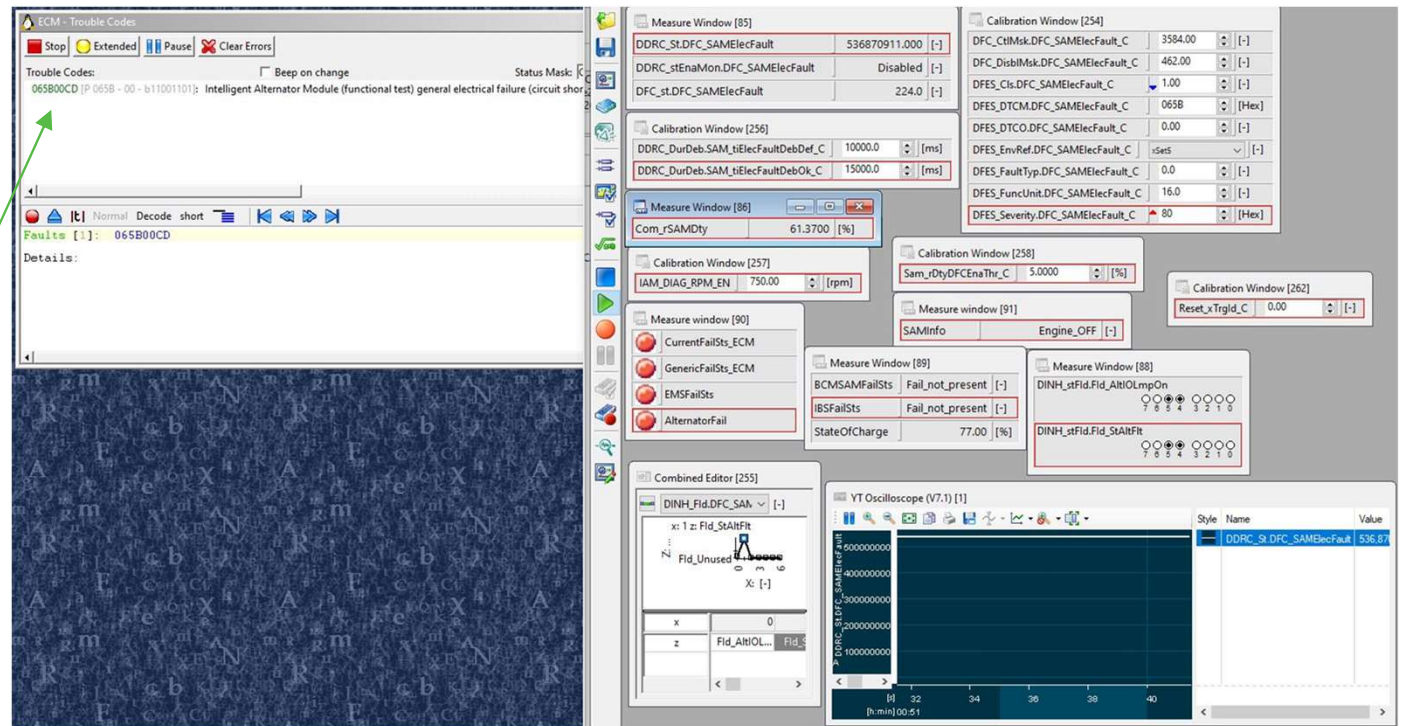


DTC test procedure 4/13

Perform a key off and wait that the ECM will turn off (power latch time)

Perform a key on and check DIAnalyzer:

Status byte has to be **CD** (if enabling conditions are not met and fault can't be tested in key on) or in **8F** (fault tested in key on and still present)



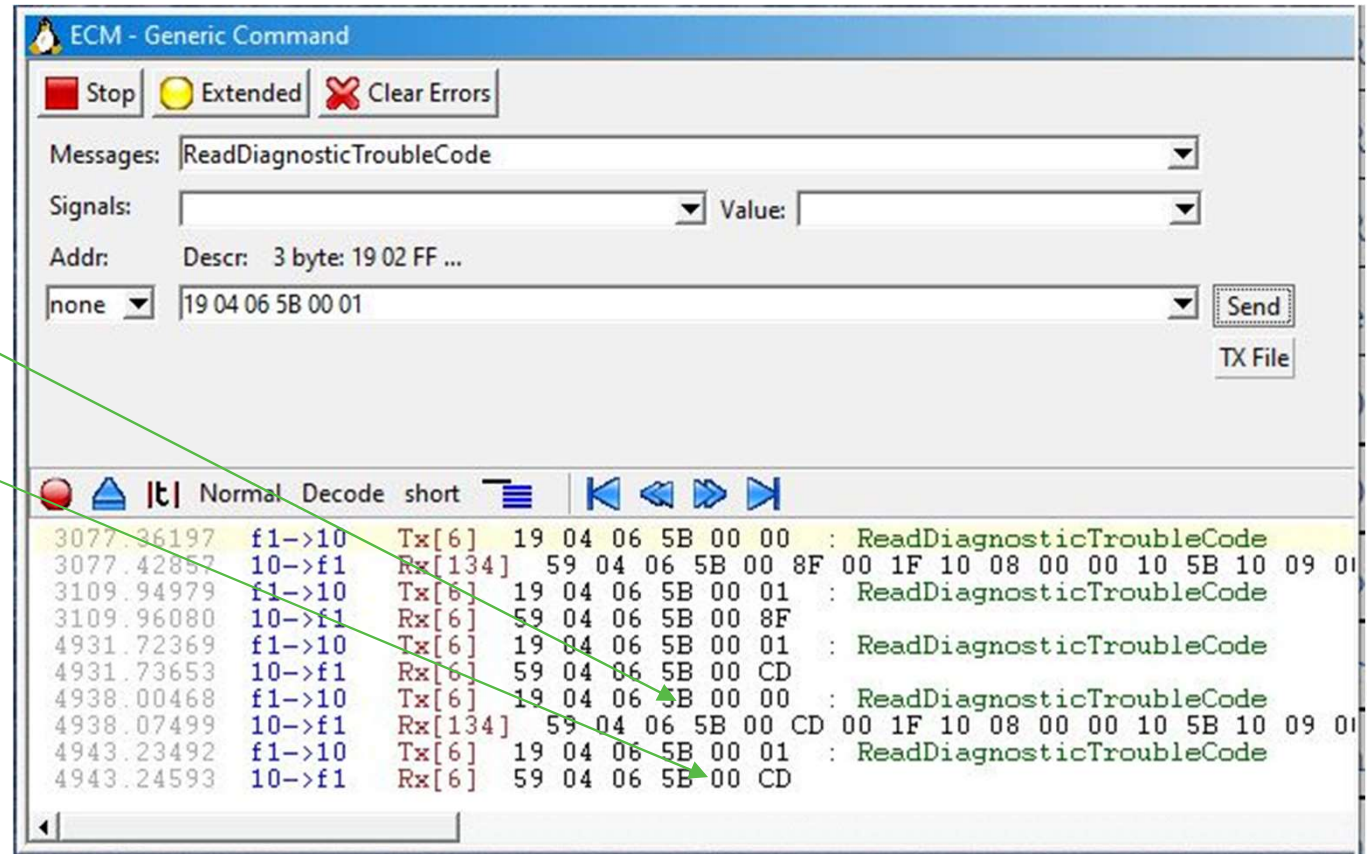
DTC test procedure 5/13

With service \$19 check that there is only 1 snapshot in the error memory (since it is the first time the DTC has been stored):

Send 19 04 06 5B 00 00 and check that snapshot is populated

Send 19 04 06 5B 00 01 and check that snapshot is void

Since the fault was never removed and never checked as healed by the ECM, the presence of a second snapshot is NOT correct.



DTC test procedure 6/13

Heal the fault according to DTC table

ElectricalFault	Constant	0	0
-----------------	----------	---	---

DTC healing conditions met after the fault was already confirmed by ECM: Status Byte will be **8E**

The screenshot displays the ECM Troubleshooting interface with several windows open:

- ECM - Trouble Codes:** Shows a list of trouble codes, including 065B008E (Intelligent Alternator Module (functional test) general electrical failure (circuit short)).
- Measure Window [85]:** Displays various parameters such as DDRC_StDfC_SAMElecFault (536870911.000), DDRC_StEnaMon.DfC_SAMElecFault (Enabled), and DfC_StDfC_SAMElecFault (232.0).
- Calibration Window [254]:** Shows calibration parameters like DfC_CtlMsk.DfC_SAMElecFault_C (3584.00), DfC_DisblMsk.DfC_SAMElecFault_C (462.00), and DfES_Ch.DfC_SAMElecFault_C (1.00).
- Calibration Window [256]:** Displays parameters like DDRC_DurDeb.SAM_tElecFaultDebDef_C (10000.0) and DDRC_DurDeb.SAM_tElecFaultDebOk_C (15000.0).
- Calibration Window [257]:** Shows parameters like IAM_DIAG_RPM_EN (750.00) and SAMInfo (Steady_State).
- Calibration Window [258]:** Displays parameters like Sam_rDyDfCEnaThr_C (5.0000) and SAMInfo (Steady_State).
- Calibration Window [262]:** Shows parameters like Reset_xTrgid_C (0.00) and SAMInfo (Steady_State).
- Measure window [90]:** Displays parameters like CurrentFailSts_ECM, GenericFailSts_ECM, EMSFailSts, and AlternatorFail.
- Measure Window [89]:** Shows parameters like BCMsAMFailSts (Fail_not_present), IBSFailSts (Fail_not_present), and StateOfCharge (77.00).
- Measure Window [88]:** Displays parameters like DINH_StFid.Fid_AltIOLmpOn (7 8 5 4 3 2 1 0) and DINH_StFid.Fid_StAIfr (7 8 5 4 3 2 1 0).
- Combined Editor [255]:** Shows a table with columns x, z, and Fid_AltIOLmpOn.
- YT Oscilloscope (V7.1) [1]:** Displays a waveform plot.

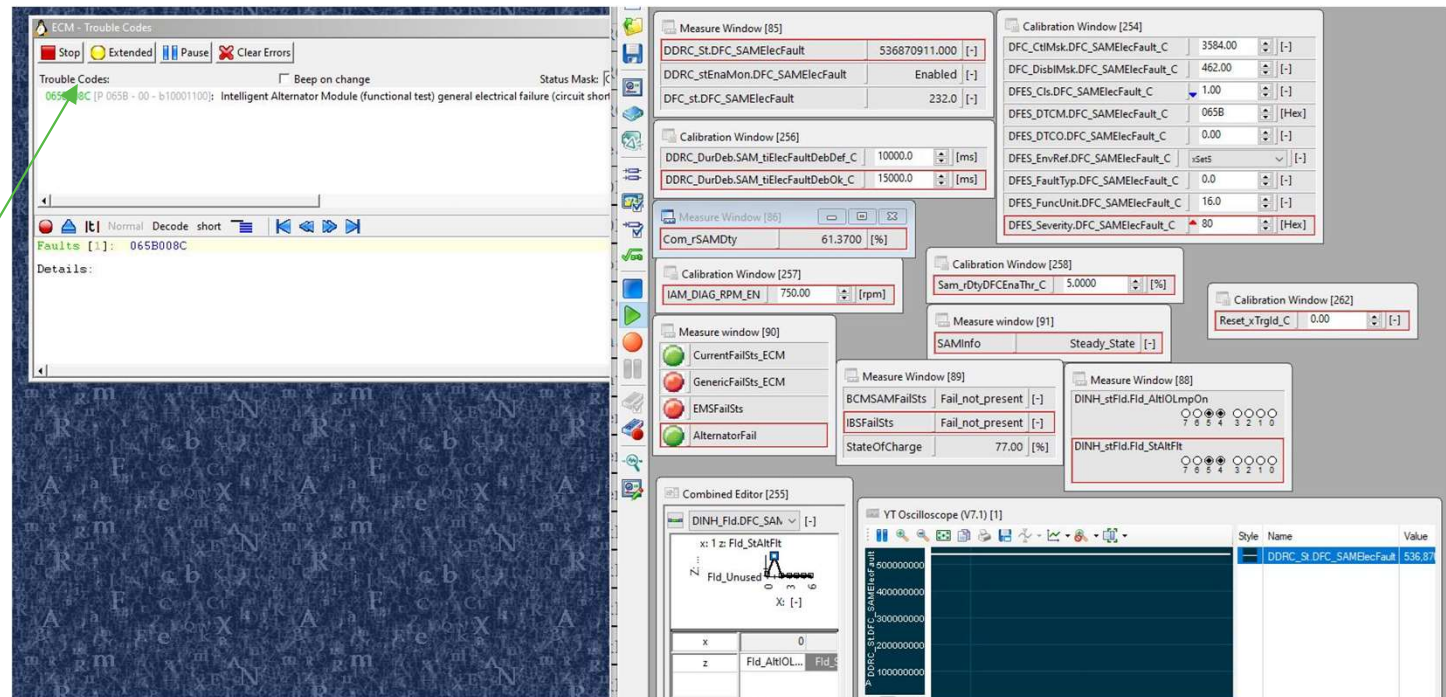
Style	Name	Value
DDRC_StDfC_SAMElecFault	536870911.000	

DTC test procedure 7/13

Perform a key off and wait that the ECM will turn off (power latch time)

Perform a key on and go in Er and check DIAnalyzer

DTC healing conditions met,
fault not present in this cycle:
Status Byte will be **8C**



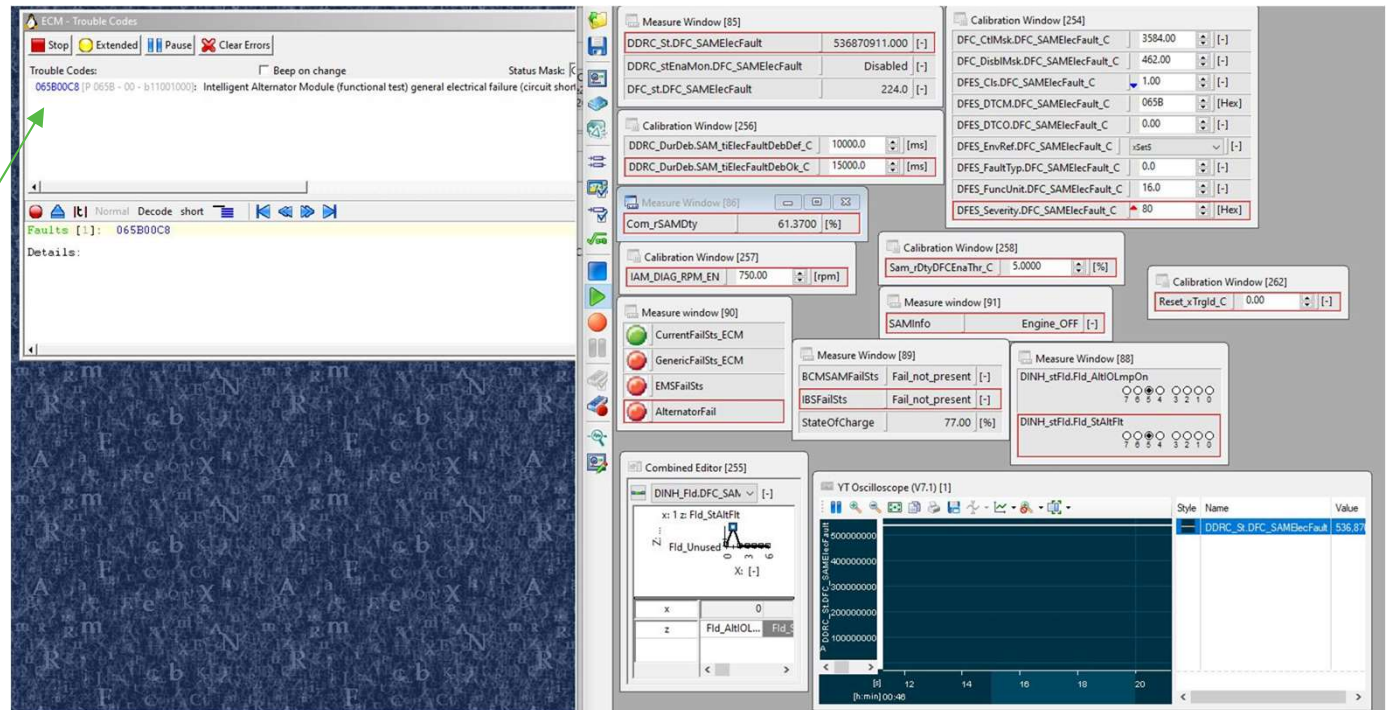
DTC test procedure 8/13

Perform a key off and DON'T wait that the ECM will turn off (power latch time)

Perform a key on and check DIAnalyzer

Alternate waiting for PL and Not waiting for PL to cover more initialization conditions of the software

DTC healing conditions not met in key on, fault not present in this cycle: Status Byte will be C8

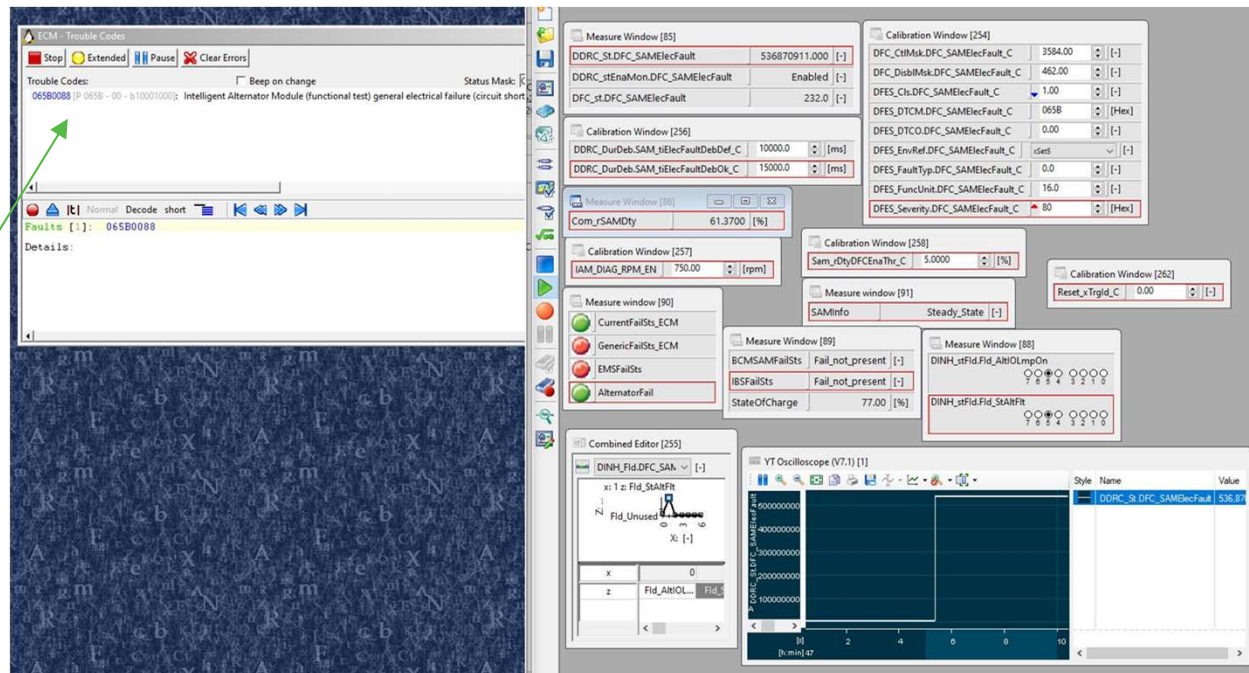




DTC test procedure 9/13

Go in Engine running and check DIAnalyzer

DTC not present in this cycle: Status Byte will be 88



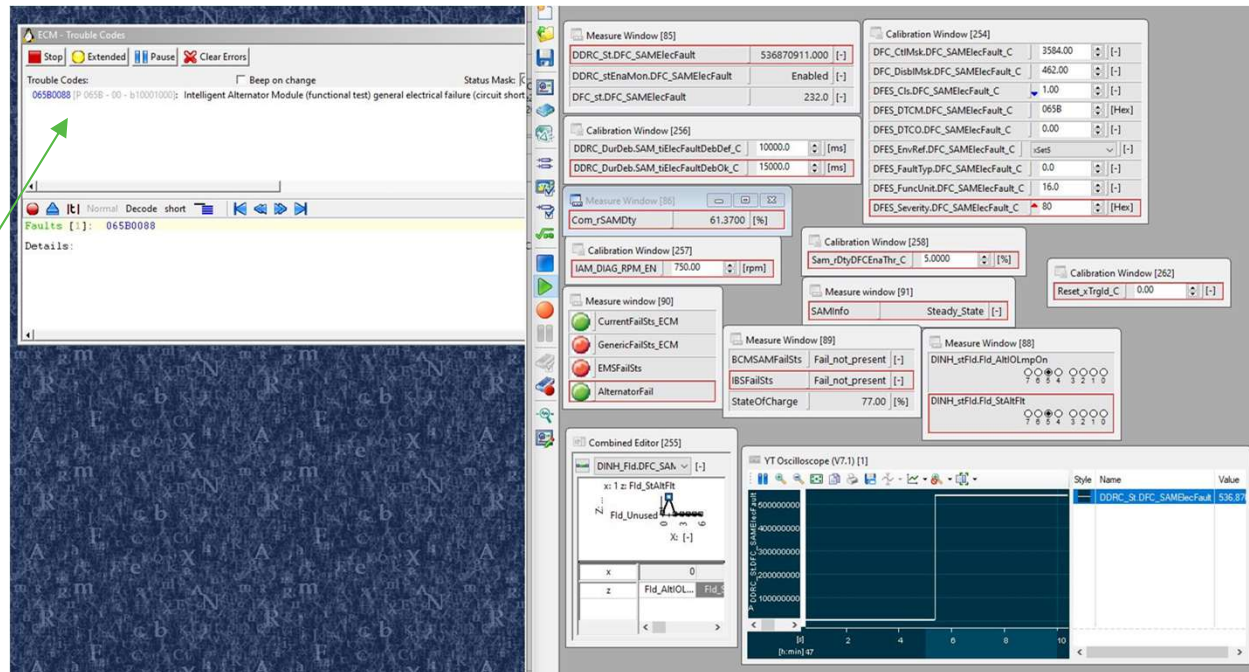
DTC test procedure 10/13

Perform a key off and DON'T wait that the ECM will turn off (power latch time)

Perform a key on and Go in Engine running and check DIAnalyzer

DTC not present in this cycle: Status Byte will be 88

For two monitoring cycles it will be 88!

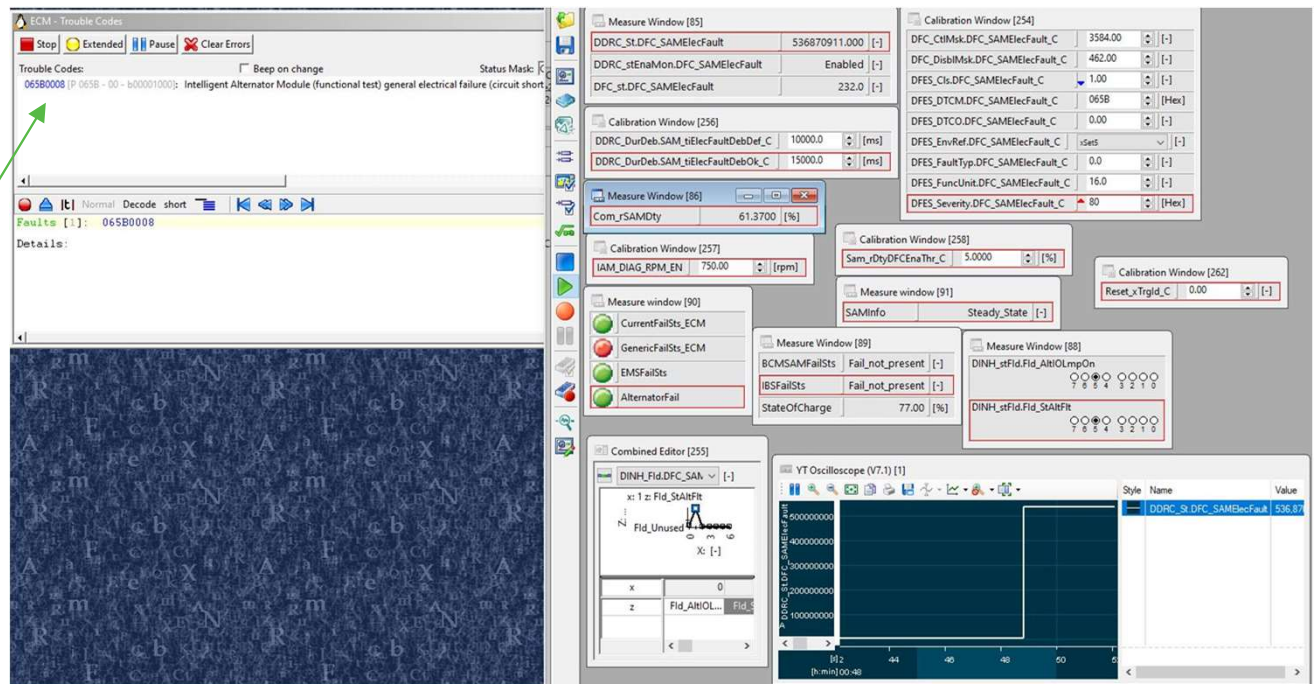


DTC test procedure 11/13

Perform a key off and DON'T wait that the ECM will turn off (power latch time)

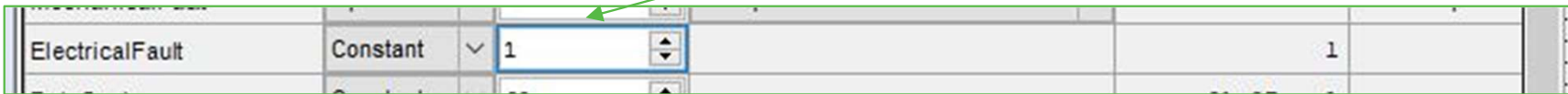
Perform a key on and Go in Engine running and check DIAnalyzer

DTC not present in this cycle: Status Byte will be 08



Perform a key off and DON'T wait that the ECM will turn off (power latch time)

Perform a key on and Go in Engine running and **set again** the fault



With service \$19 check that there are 2 snapshots in the error memory (since it is the second time the DTC has been stored):

Send 19 04 06 5B 00 00 and check that snapshot is populated

Send 19 04 06 5B 00 01 and check that snapshot is populated.

Check coherence of second snapshot too!

With service \$19 check that there are 2 snapshots in the error memory (since it is the second time the DTC has been stored):

Send 19 04 06 5B 00 00 and check that snapshot is populated

Send 19 04 06 5B 00 01 and check that snapshot is populated.

Check coherence of second snapshot too!

With service \$19 check that there are 2 snapshots in the error memory (since it is the second time the DTC has been stored):

Send 19 04 06 5B 00 00 and check that snapshot is populated

Send 19 04 06 5B 00 01 and check that snapshot is populated.

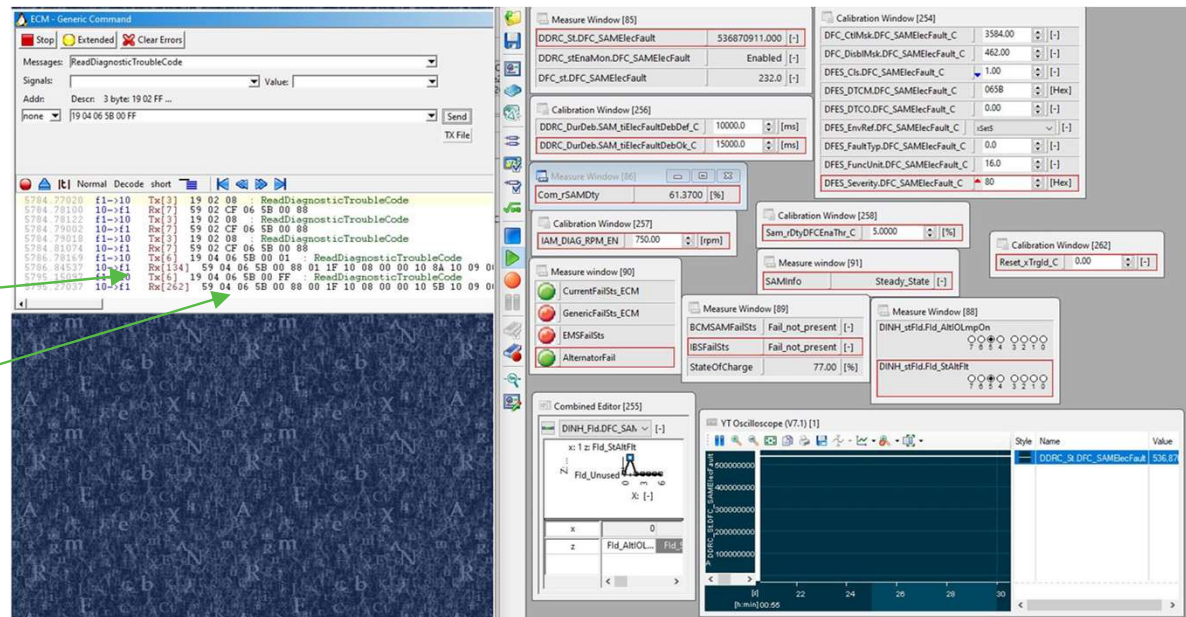
Check coherence of second snapshot too!

With service \$19 check that there are 2 snapshots in the error memory (since it is the second time the DTC has been stored):

Send 19 04 06 5B 00 00 and check that snapshot is populated

Send 19 04 06 5B 00 01 and check that snapshot is populated.

Check coherence of second snapshot too!



DTC test procedure 13/13

Save the acquisitions from INCA and from DIAnalyzer.

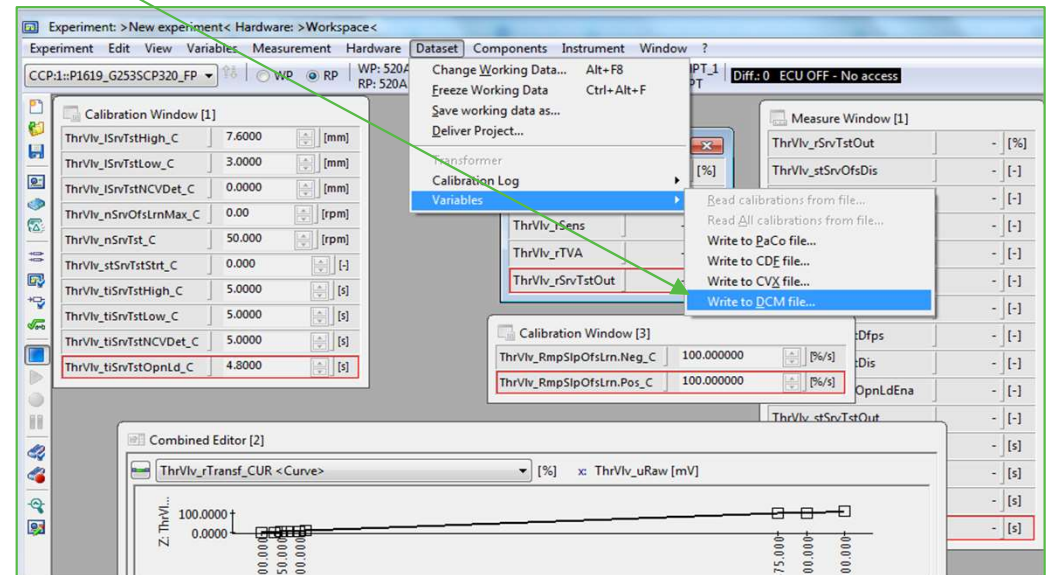
Create a DCM file of all the calibrations used by your experiment

INCA: DFC_APPPlaUsBrk_P2299_520AMBI50x1_20200910.dat
DIAnalyzer: DFC_APPPlaUsBrk_P2299_520AMBI50x1_20200910.log
DCM: DFC_APPPlaUsBrk_P2299_520AMBI50x1_20200910.dcm

Write in the report file the results and the name of the acquisitions

Open the point in the OPL list if you have found an issue and send an email to the OPL mailing list

Each issue has to have its own point, don't mix issues in a single OPL item



Object of the OPL emails:

SoftwareLineProject_FileNameDVPM_Dataset_#XXXX: Short description

e.g. for G-Line 520 1.6l DS Eu6D-Final with dataset 520AMBI52X1:

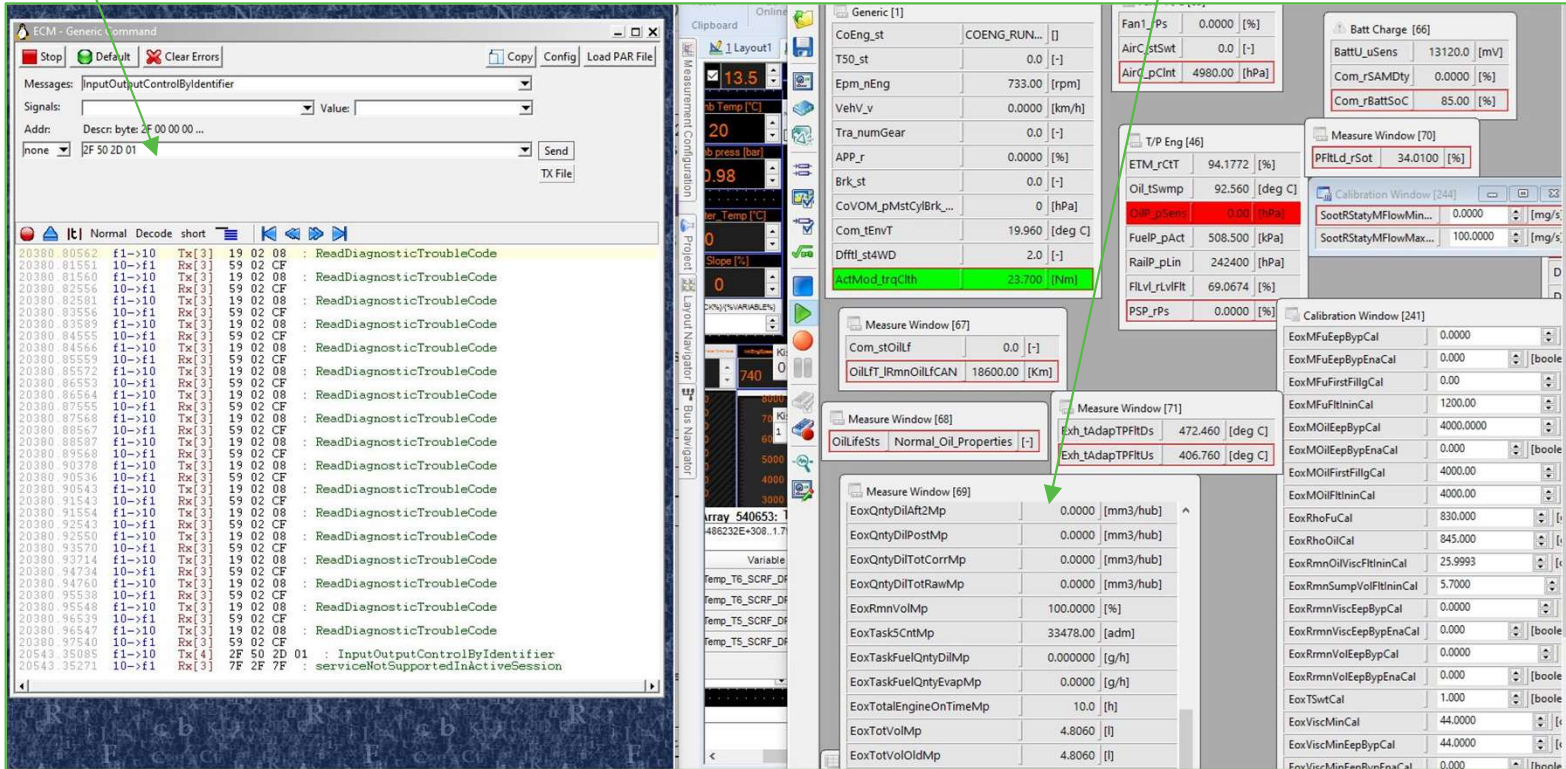
P1619_DVPM_2.1_FamB_E6dFinal_520AMBI52X1_#1272: Speed Limiter - 2 key cycle ISSUE

Start IOLI/Routine in all diagnostic sessions both in physical and in functional addressing

Acquisition prepared with all measuring points and all details of actuator under test

Compare with CDD the results and report if there are mismatches

Try to send positive and negative responses requests in Power On and in Engine running vehicle conditions



IOLI/Routine test procedure 3/5

Repeat the tests in the Functional Addression

Physical addressing: Addr=none
Functional addressing: Addr=ALL

In functional addressing the Diagnostic request will be answered by all ECUs

If no ECUs will answer, then Rx Timeout will happen

The screenshot displays a diagnostic tool interface with multiple windows. The main window shows a list of CAN bus messages. A green arrow points from the text 'Repeat the tests in the Functional Addression' to the 'Addr' field in the 'Messages' section, which is set to 'ALL'. Another green arrow points from the text 'If no ECUs will answer, then Rx Timeout will happen' to a message in the list with the description 'Error -2011: Rx Timeout'. The interface also includes several data windows on the right side, such as 'CoEng_st', 'Fan1_rPs', 'Batt Charge', 'T/P Eng', 'OilLifeSts', 'Normal_Oil_Properties', 'Exh_tAdapTPFRIDs', 'Exh_tAdapTPFRUs', 'EoxQtyDilAft2Mp', 'EoxQtyDilPostMp', 'EoxQtyDilTotCorrMp', 'EoxQtyDilTotRawMp', 'EoxRmnVolMp', 'EoxTaskCntMp', 'EoxTaskFuelQtyDilMp', 'EoxTaskFuelQtyEvapMp', 'EoxTotalEngineOnTimeMp', 'EoxTotVolMp', 'EoxTotVolOldMp', 'Fan1_rPs', 'AirC_stSwt', 'AirC_pCInt', 'T/P Eng', 'ETM_rCtT', 'OilTsWmp', 'FuelP_pAct', 'RailP_pLin', 'FILvL_tVIFRt', 'PSP_rPs', 'Batt Charge', 'Com_rSAMDTy', 'Com_rBattSoC', 'Measure Window [70]', 'Calibration Window [244]', 'Measure Window [67]', 'Measure Window [68]', 'Measure Window [71]', 'Measure Window [69]', and 'Calibration Window [241]'. The 'Error -2011: Rx Timeout' message is highlighted in the main list.

IOLI/Routine test procedure 2/5

Engine running condition is excluded by all IOLIs as a general rule, check the CDD for the specific vehicle conditions required

Test all positive and negative responses and check that all of them are reported in CDD

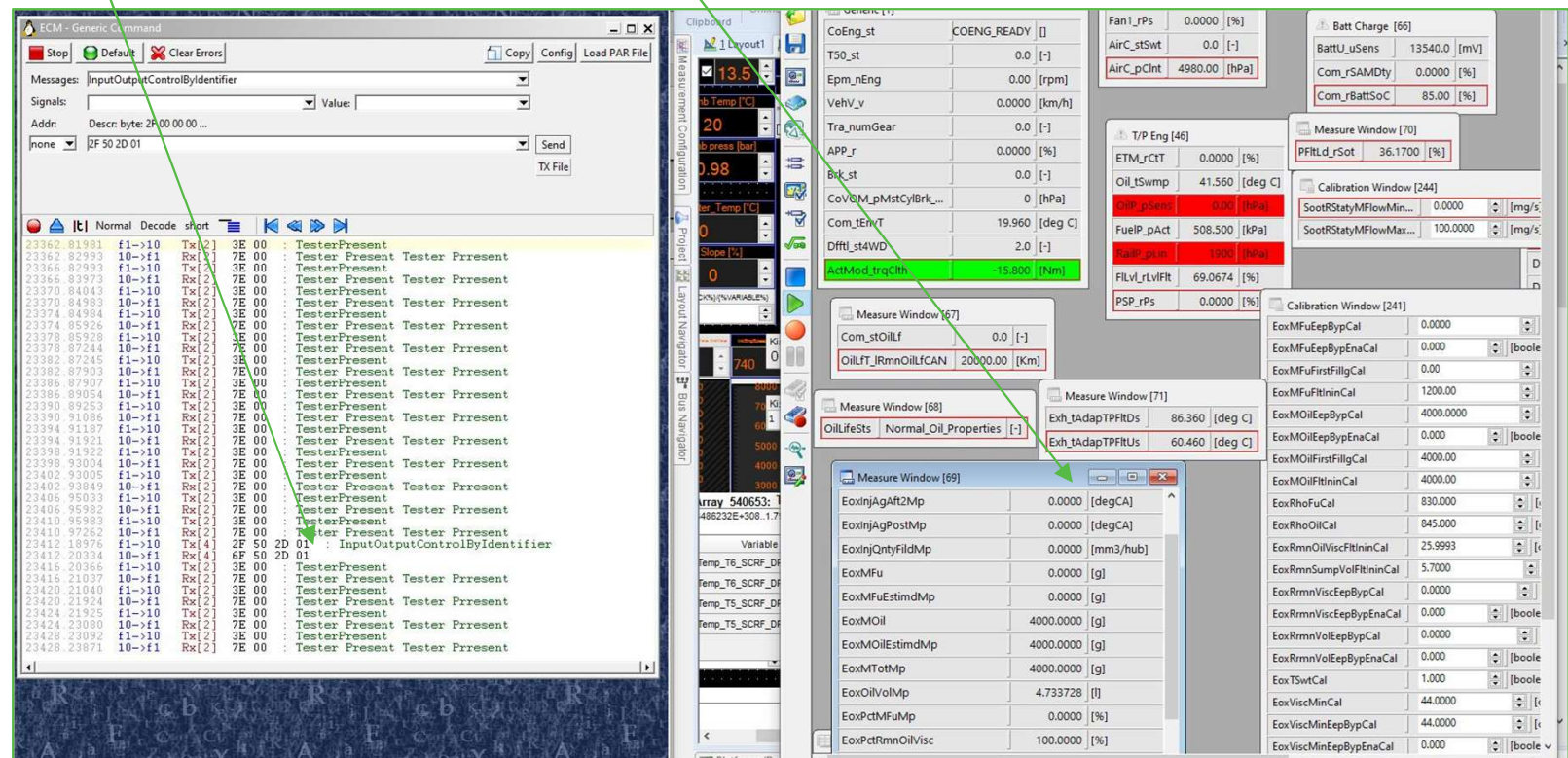
The screenshot displays a diagnostic software interface with multiple windows. The main window shows a list of diagnostic trouble codes (DTCs) and their associated data. The DTCs are listed in a table with columns for time, address, data, and description. The descriptions include 'ReadDiagnosticTroubleCode', 'serviceNotSupportedInActiveSession', 'StartDiagnosticSession', 'ExtendedDiagnostic Start', 'InputOutputControlByIdentifier', 'ConditionsNotCorrect', 'TesterPresent', and 'Tester Present Tester Present'.

Other windows show various vehicle parameters and measurements, including:

- CoEng_st, COENG_RUN...
- T50_st, 0.0 [-]
- Epm_nEng, 810.50 [rpm]
- Veh_v, 0.0000 [km/h]
- Tra_numGear, 0.0 [-]
- APP_r, 0.0000 [%]
- Brk_st, 0.0 [-]
- CoVOM_pMstCylBrk..., 0 [hPa]
- Com_EnvT, 19.960 [deg C]
- Dfflt_st4WD, 2.0 [-]
- ActMod_trqClth, -3.400 [Nm]
- Measure Window [67]: Com_stOilF, 0.0 [-]; OilFt_IRmOilFCAN, 18600.00 [Km]
- Measure Window [68]: OilLifeSts, Normal_Oil_Properties [-]
- Measure Window [71]: Exh_tAdapTPFRDs, 467.760 [deg C]; Exh_tAdapTPFRUS, 291.260 [deg C]
- Measure Window [69]: EoxQtyDilAft2Mp, 0.0000 [mm3/hub]; EoxQtyDilPostMp, 0.0000 [mm3/hub]; EoxQtyDilTotCorrMp, 0.0000 [mm3/hub]; EoxQtyDilTotRawMp, 0.0000 [mm3/hub]; EoxRmnVolMp, 100.0000 [%]; EoxTask5CntMp, 36633.00 [adm]; EoxTaskFuelQtyDilMp, 0.000000 [g/h]; EoxTaskFuelQtyEvapMp, 0.0000 [g/h]; EoxTotalEngineOnTimeMp, 10.0 [h]; EoxTotVolMp, 4.8060 [l]; EoxTotVolOldMp, 4.8060 [l]
- Fan1_rPs, 0.0000 [%]; AirC_stSwT, 0.0 [-]; AirC_pClnt, 4980.00 [hPa]; Batt_uSens, 13340.0 [mV]; Com_rSAMDty, 0.0000 [%]; Com_rBattSoC, 85.00 [%]; T/P Eng [46]: ETM_rCT, 100.0000 [%]; OilTswmp, 92.460 [deg C]; FuelIP_pAct, 508.500 [kPa]; RailIP_pLin, 248200 [hPa]; FILVL_rLVIFt, 69.0674 [%]; PSP_rPs, 0.0000 [%]; Measure Window [70]: PFFtLd_rSot, 36.7800 [%]; Calibration Window [244]: SootRStatyMFlowMin..., 0.0000 [mg/s]; SootRStatyMFlowMax..., 100.0000 [mg/s]; Calibration Window [241]: EoxMFuEepBypCal, 0.0000; EoxMFuEepBypEnaCal, 0.0000; EoxMFuFirstFillgCal, 0.00; EoxMFuFitinCal, 1200.00; EoxMOilEepBypCal, 4000.0000; EoxMOilEepBypEnaCal, 0.0000; EoxMOilFirstFillgCal, 4000.00; EoxMOiFitinCal, 4000.00; EoxRhoFuCal, 830.0000; EoxRhoOilCal, 845.0000; EoxRmnOilViscFitinCal, 25.9993; EoxRmnSumpVolFitinCal, 5.7000; EoxRmnViscEepBypCal, 0.0000; EoxRmnViscEepBypEnaCal, 0.0000; EoxRmnVolEepBypCal, 0.0000; EoxRmnVolEepBypEnaCal, 0.0000; EoxTswtCal, 1.000; EoxViscMinCal, 44.0000; EoxViscMinEepBypCal, 44.0000; EoxViscMinEepBypEnaCal, 0.0000

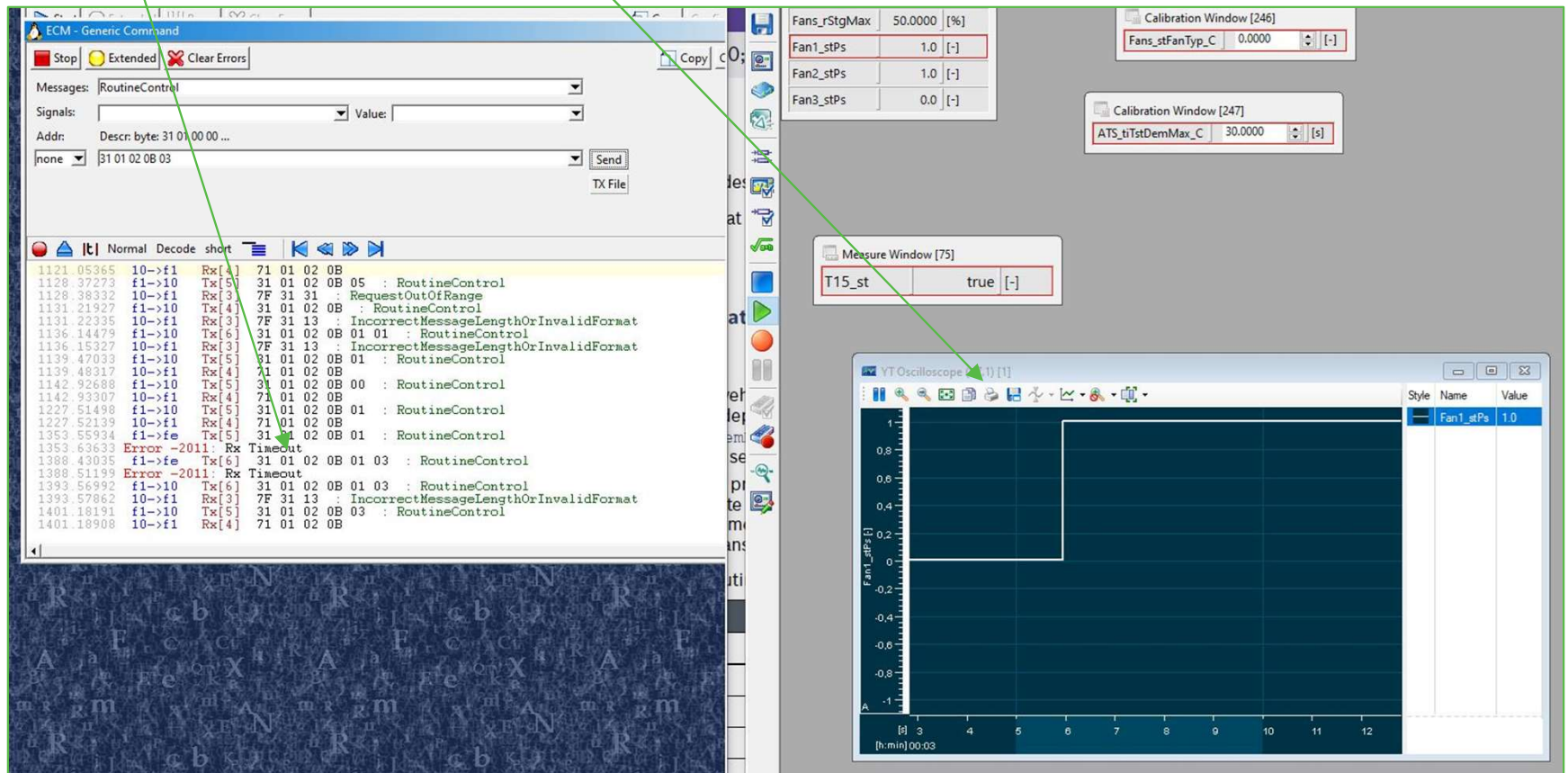
IOLI/Routine test procedure 4/5

Check the that when IOLI answers with positive response, what is requested by CDD will happens: for the example reported, the reset of internal parameters



IOLI/Routine test procedure 5/5

Check the execution time for both IOLI and Routine accordingly with the CDD requests



RDI test procedure 1/3

Check internal to physical conversion formulas

$\text{phys} = \text{fac_sw} \times \text{int} + \text{offset_sw}$
 $900 \text{ rpm} = 0.50 \times 1800 + 0$

For the physical to internal conversion formula, the software formulas have to be used (search them in SW manual)

The screenshot displays the RDI test software interface. On the left, the 'ECM' window shows a list of messages, including 'ReadDataByIdentifier' and 'RDI_1000 Engine speed Read'. The 'Engine speed' is listed as 900.00 rpm. In the center, a 'Clipboard' window shows a list of variables, including 'DFES_numDFC' and 'DFES_stChk'. On the right, a 'Calibration Window' shows various parameters, including 'SyC_tiPostDrv' (0.0 [ms]), 'DFES_DisblMsk_DFC_SyncAPP_C' (132.00), and 'Epm_nEng' (900.00 [rpm]). A 'Measure Window' shows 'Epm_nEng' (1800 [dec]). The bottom right corner shows a 'Calibration Window' with 'A_tICANARShutOff_C' (180000.0 [ms]).

RDI test procedure 2/3

Check for at least 3 measuring values for the linear RDIs, check every bit for the encoded bits ones

D&C conversion formula is used to convert the hex value to the physical value in the diagnostic tester. It NOT the internal representation of the RDI

HEX value = 2E E0
DEC value = 12000
D&C conversion formula:
FAC=1, DEN=4, OFS=0

Phys = (FAC x DEC)/DEN + OFS
3000 rpm = (1 x 12000)/4 + 0

The screenshot displays a diagnostic tool interface with several windows. The main window shows a list of messages, including 'ReadDataByIdentifier' and 'RDI_1000 Engine speed Read'. The 'Engine speed' is displayed as 3000 rpm. A 'Calibration Window' is open, showing the conversion formula: $Phys = (FAC \times DEC) / DEN + OFS$. The values are FAC=1, DEC=12000, DEN=4, and OFS=0, resulting in 3000 rpm. The 'Measure Window' shows the physical value of 3000.00 [rpm]. The 'Calibration Window' also shows the conversion formula and the resulting physical value.

DFES_numDFC	DFC_SyncApp	DFES_stChk	DFES_stEntry
DFES_numDFC[1]	DFC_SyncApp	DFES_stChk[1]	DFES_stEntry[1]
DFES_numDFC[2]	DFC_SyncApp	DFES_stChk[2]	DFES_stEntry[2]
DFES_numDFC[3]	NoEntry	DFES_stChk[3]	DFES_stEntry[3]
DFES_numDFC[4]	NoEntry	DFES_stChk[4]	DFES_stEntry[4]
DFES_numDFC[5]	NoEntry	DFES_stChk[5]	DFES_stEntry[5]
DFES_numDFC[6]	NoEntry	DFES_stChk[6]	DFES_stEntry[6]
DFES_numDFC[7]	NoEntry	DFES_stChk[7]	DFES_stEntry[7]
DFES_numDFC[8]	NoEntry	DFES_stChk[8]	DFES_stEntry[8]
DFES_numDFC[9]	NoEntry	DFES_stChk[9]	DFES_stEntry[9]

RDI test procedure 3/3

Check all positive and negative responses reported in CDD. Check both in physical and functional addressing and in all diagnostic sessions

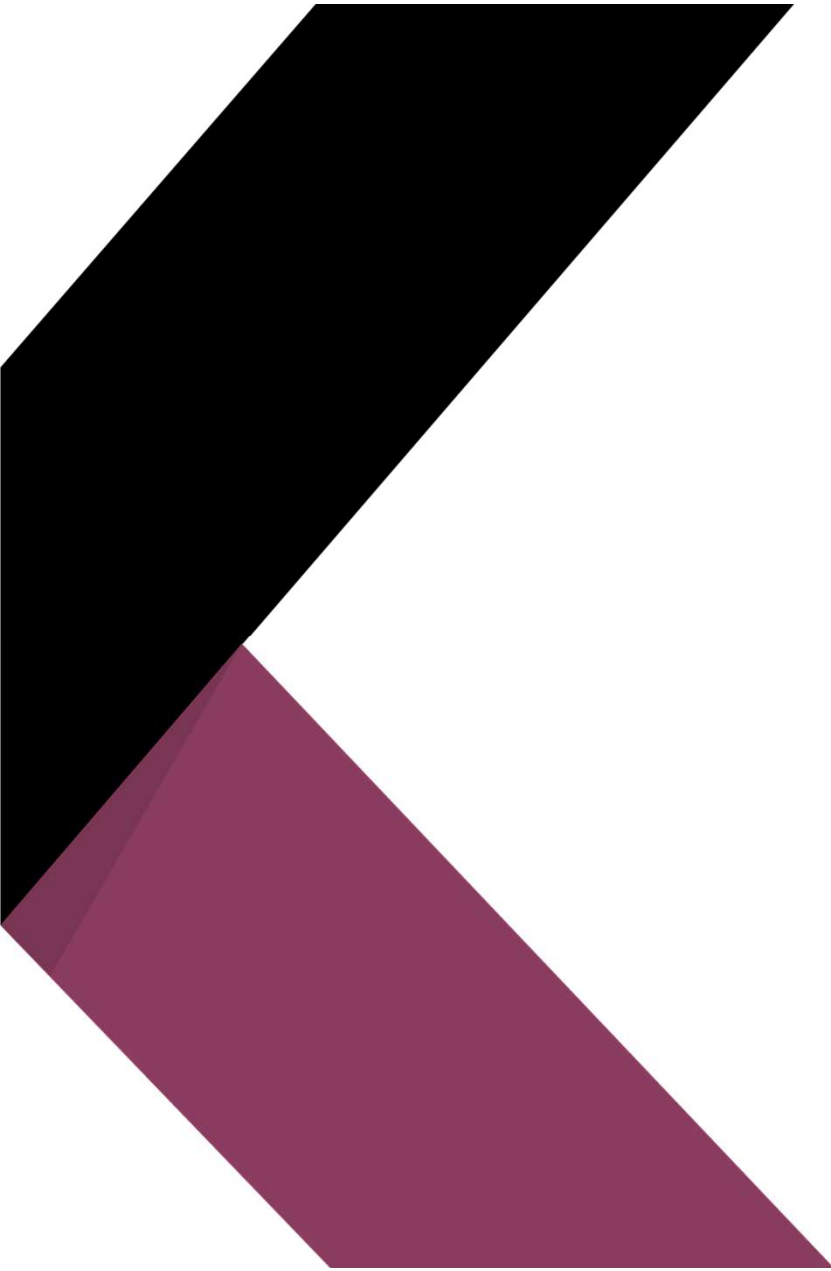
Perform all maneuvers to fully test all conditions for the non linear RDIs

Report if there are mismatching conditions

The screenshot displays the ECM - Generic Command interface. The main window shows a list of diagnostic messages with columns for time, address, data, and description. A green arrow points to a message at 439.74351: Rx[3] 7F 10 78 : RequestCorrectlyReceived-ResponsePending. To the right, there are several calibration windows (e.g., Calibration Window [223], Calibration Window [5], Calibration Window [234], Calibration Window [240], Calibration Window [245], Calibration Window [239]) and measure windows (e.g., Measure Window [43], Measure Window [72]). The interface also includes a status bar at the bottom and a sidebar with various icons.



Diagnosis Process



CONTACTS

Via E. Gianturco, 23
(presso complesso Mecfond)
80146 Napoli

www.kineton.it

